

Compact Wisdom at Small Scale: Small Language Models as Cultural Assistants

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Abstract—Large language models (LLMs) offer strong reasoning and generation, but their training and inference costs limit deployment in low-resource settings. Small language models (SLMs) in the 100M–1B parameter range promise frugal inference, yet it remains unclear whether they can support culturally grounded assistance with high faithfulness and explicit attribution. We study this question using the *Thirukkural*, a canonical Tamil ethical text of 1,330 aphorisms with bilingual translations and commentaries. We curate a bilingual (Tamil–English) instruction corpus anchored at the verse level and build a retrieval-augmented generation (RAG) pipeline that enforces grounding via evidence blocks and citation constraints. Our retriever uses hybrid sparse–dense retrieval with bilingual reranking, and we introduce hierarchical indexing strategies aligned to the *Paal–Iyal–Adhigaram* structure, including a clustered Adhigaram approach backed by a deterministic knowledge graph. We evaluate Gemma-3 270M and a 1B variant, with and without LoRA-based parameter-efficient fine-tuning. Results show that retrieval structure is the dominant factor for both grounding and generation quality: moving from flat Kural-level indexing to clustered Adhigaram-level retrieval substantially improves recall, ranking stability, and downstream semantic fidelity. Under structured retrieval, SLMs approach larger-model baselines in semantic quality (e.g., BERTScore-F1 ≥ 0.80) while remaining feasible on commodity hardware. Overall, the findings support a retrieval-first recipe for deployable cultural assistants over bounded canonical corpora, where hierarchical evidence selection and lightweight adaptation can offset limited model capacity.

Index Terms—Thirukkural, Small Language Models, Retrieval-Augmented Generation, Cultural Grounding, Hierarchical Indexing, Knowledge Graphs, Parameter-Efficient Fine-Tuning

I. INTRODUCTION

Large Language Models (LLMs) demonstrate strong general-purpose reasoning and generation capabilities, but their substantial training cost, memory footprint, and energy consumption limit deployment in schools, mobile devices, and other low-resource environments [1]. These constraints have renewed interest in Small Language Models (SLMs) [2], which trade broad coverage for efficiency, lower fine-tuning and inference cost, and ease of deployment. SLMs are particularly

effective when task scope is narrow, domain boundaries are well-defined, and authoritative sources are stable [3]. Under such conditions, retrieval-augmented generation (RAG) can compensate for limited parametric knowledge by grounding outputs in explicit external corpora [4].

RAG also plays a critical role in enforcing *grounding*, enabling models to retrieve canonical evidence and condition generation on retrieved text rather than latent associations alone. This is especially important for *cultural or civilizational assistants*—AI systems intended to provide guidance rooted in specific textual traditions or ethical canons. Such systems impose two stringent requirements: (i) *faithfulness*, ensuring responses do not deviate from source texts, and (ii) *transparent attribution*, requiring explicit citation of authoritative evidence. Whether these constraints require large-scale models, or can instead be satisfied by smaller models aligned through retrieval and task-specific adaptation, remains an open empirical question [2].

We study this question using the *Thirukkural* [5], a compact yet philosophically dense Tamil corpus consisting of 1,330 aphoristic couplets. The text is organized into a stable hierarchical structure of *Paal* (books), *Iyal* (chapters), and *Adhigaram* (sections), each addressing concrete aspects of ethical conduct, governance, and human relationships. Its finite size, canonical stability, and clear thematic segmentation make it particularly well suited for retrieval-based grounding, allowing models to rely on explicit textual evidence rather than extensive parametric memorization.

We formulate a practical task: **given a moral or situational query, retrieve the most relevant Kurals and generate a brief explanation explicitly grounded in the retrieved verse and its commentary**. This setup allows us to isolate whether SLMs, when coupled with retrieval, can deliver faithful, interpretable, and succinct responses under strict resource constraints, and to study the marginal benefits of lightweight task-specific adaptation.

Our contributions are three-fold and centered on the following research question:

RQ1: *To what extent can small language models, when augmented with retrieval and lightweight task-specific adaptation [6], achieve faithful and attributable generation comparable to large language models on grounded ethical reasoning tasks?*

First, we curate a bilingual (Tamil–English) instruction and supervision corpus derived from the *Thirukkural*, explicitly designed to separate retrieval accuracy from generation faithfulness. Second, we propose a retrieval pipeline tailored to compact canonical corpora using sparse lexical retrieval (BM25), along with post hoc grounding checks that verify verse-level alignment between retrieved evidence and generated explanations. Third, we systematically compare (i) base SLMs without retrieval, (ii) SLMs with RAG, and (iii) SLMs with RAG and LoRA-based adaptation, using large language models as reference baselines, isolating when scale is necessary and when grounded, task-aligned small models suffice.

II. RELATED WORK

a) Small Language Models and Parameter-Efficient Adaptation: Large language models achieve strong general-purpose performance but remain expensive to train and deploy, motivating renewed interest in small language models (SLMs) with hundreds of millions to a few billion parameters [1], [7], [8]. Prior work primarily frames SLMs as compressed or distilled variants of LLMs, optimized for latency and memory efficiency. Parameter-Efficient Fine-Tuning (PEFT) methods, particularly Low-Rank Adaptation (LoRA) [9], have further reduced adaptation costs by training a small number of additional parameters while freezing the backbone model. Extensions such as LoRA-SP, MELoRA, and RA-LoRA demonstrate that effective task adaptation is possible using only a small fraction of model capacity, especially in low-resource or multilingual settings [10]. However, existing PEFT work largely targets generic downstream tasks such as classification or translation. In contrast, our work positions SLMs as *faithful cultural assistants*, where correctness is defined not by fluency or task accuracy alone, but by explicit grounding and attribution to canonical sources.

b) Retrieval-Augmented Generation and Structured Grounding: Retrieval-Augmented Generation (RAG) was introduced to mitigate hallucination and parametric knowledge limits by conditioning generation on retrieved documents [11]. Subsequent work has explored dense retrieval, hybrid sparse–dense methods, and multi-stage retrieval pipelines [12]. Most RAG systems, however, rely on flat document chunking and embedding-based similarity, which is poorly suited for compact, aphoristic, and philosophically dense corpora. Recent structured variants, including hierarchical and multi-hop retrieval, attempt to address this limitation [13]. Graph-based approaches further model inter-document relationships but typically depend on large models and expensive inference [14]. Our approach adopts a lightweight hierarchical retrieval strategy aligned with the inherent structure of the *Thirukkural*, enabling precise verse-level

grounding without the computational overhead of graph construction or large-model reasoning. This design choice allows retrieval structure—not model scale—to carry the burden of interpretive coherence.

c) Grounded Evaluation and LLM-as-a-Judge: Evaluating grounded generation remains challenging, particularly for culturally sensitive or interpretive tasks. Human evaluation is the gold standard but does not scale, motivating hybrid evaluation pipelines that combine human annotation with LLM-based judging [15], [16]. Prior work shows that calibrated LLM judges using ordinal rating scales can achieve strong agreement with human preferences when prompts are carefully designed [17], [18]. Recent studies emphasize that evaluation protocols must explicitly separate retrieval quality from generation quality and must test generalization beyond memorized training examples [19]. In this work, we follow these principles by combining multi-annotator human checks with a calibrated LLM-as-a-judge and enforcing dataset splits that hold out entire subsets of verses, enabling a controlled assessment of faithfulness, attribution, and retrieval-driven generalization.

III. METHODOLOGY

To answer RQ1, we use a controlled setup: the corpus, query set, prompts, and evaluation protocol are fixed, while the model configuration is varied across (i) standalone inference, (ii) RAG inference, and (iii) RAG with parameter-efficient fine-tuning [11], [20]. This isolates the contribution of retrieval and lightweight adaptation under identical conditions.

A. Task Definition

Given a natural language query q describing a moral dilemma or reflective situation, the system must (1) retrieve relevant Kural(s) and (2) generate a short English explanation grounded in the retrieved evidence. Let $\mathcal{C} = \{(\nu_i, x_i)\}_{i=1}^{1330}$ denote the corpus, where ν_i is the Tamil verse and x_i is an aligned English translation/explanation. The system outputs $(\hat{V}, e^*)$, where $\hat{V} \subset \mathcal{C}$ is the retrieved set and e^* is an English explanation conditioned only on $\hat{V}$. Outputs that are uncited, cite non-retrieved Kurals, or introduce claims unsupported by $\hat{V}$ are treated as failures.

B. Corpus and Supervision Data

We construct a verse-level bilingual resource over all 1,330 Kurals, containing: (i) Tamil verse, (ii) English transliteration, and (iii) English translation/explanation from public authoritative editions (Tamil+transliteration from Kaggle and Project Madurai). This enables English-only generation while preserving Tamil verse citation for auditability.

a) QA construction.: We create question–Kural pairs using a mixed strategy: (a) human-authored queries collected from non-expert users to reflect everyday dilemmas, and (b) LLM-generated queries to expand linguistic and thematic coverage. For each query, an LLM proposes multiple candidate Kurals and selects the best match; we additionally generate a short reference explanation for supervision (used as a target for IFT, not as a required reasoning trace).

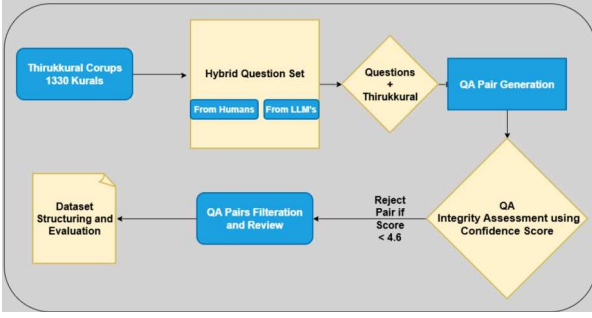


Fig. 1: Data construction and split design.

b) *Quality control and splits.*: Each query–Kural pair is validated by three annotators for semantic alignment. In parallel, an LLM-as-a-judge assigns a confidence score in [1, 5]; we retain pairs with score ≥ 4.6 and confirm reliability via spot-checking. The resulting dataset contains 1,114 distinct Kurals with coverage across all 133 *Adhigarams*. We use an 80/10/10 train/val/test split (2,193/219/219 pairs). An additional 216 Kurals are excluded from all splits to test retrieval-driven generalization (unseen during fine-tuning, accessible only via retrieval at test time). Fig. 1 summarizes the pipeline.

C. Models and Fine-Tuning

We evaluate two SLMs: Gemma-3 270M [21] and a 1B-parameter variant. All models use a unified prompt template; in RAG settings, the prompt includes an evidence block with Kural IDs, Tamil verse, transliteration, and English explanation. Generation is constrained to English.

a) *Instruction fine-tuning (IFT).*: IFT examples are formatted as (query, evidence, explanation). We optimize next-token likelihood over the assistant explanation tokens only, treating query+evidence as conditioning context [22]:

$$\mathcal{L}(\theta) = - \sum_{t \in M} \log P_{\theta}(x_t | x_{<t}). \quad (1)$$

b) *LoRA adaptation.*: We apply LoRA [9] in attention projection layers. Unless stated otherwise: $r \in \{8, 16\}$, $\alpha \in \{16, 32\}$, dropout 0.05, batch size 16–64, 3–7 epochs, learning rate 10^{-5} – 10^{-3} . Optimization uses AdamW ($\beta_1=0.9$, $\beta_2=0.95$) with weight decay 0.01, gradient clipping 1.0, label smoothing $\eta=0.1$, and bf16 precision [23]–[25]. Early stopping is based on validation loss. All fine-tuning runs use an NVIDIA A6000 (48GB).

D. Retrieval and Evidence Construction

Retrieval is performed at the verse level; each indexed item contains Kural ID, Tamil verse, transliteration, English translation/explanation, and chapter metadata.

a) *Candidate retrieval.*: We use hybrid sparse+dense retrieval: BM25 [26] over English translations/explanations and dense retrieval using BGE-M3 embeddings [27]. Top- k candidates (default $k=10$) are retrieved.

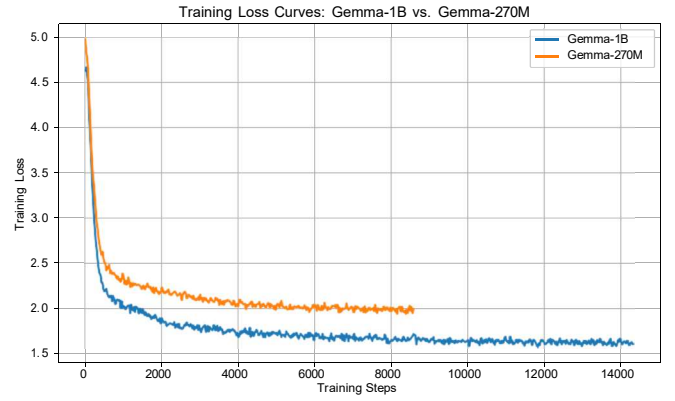


Fig. 2: Training loss.

b) *Reranking and evidence block.*: Candidates are reranked using a bilingual cross-encoder scoring (q, v) semantic alignment [28]. The final top- k Kurals form the evidence block passed verbatim to the generator.

c) *Attribution constraint.*: Outputs must cite Kural IDs. During evaluation, citing a Kural not present in the evidence block is counted as an attribution error [29].

E. Indexing Strategies

We study three indexing granularities.

a) (i) *Kural-level.*: Each Kural is an atomic retrieval unit. This maximizes verse-level precision but is sensitive to paraphrase and thematic dispersion across neighboring verses.

b) (ii) *Adhigaram-level.*: All Kurals within an *Adhigaram* are grouped as one retrieval unit by concatenating their English content, improving thematic localization at the cost of introducing extra context.

c) (iii) *Clustered Adhigaram-level.*: We build a deterministic Neo4j graph encoding the canonical hierarchy (*Paal* → *Iyal* → *Adhigaram* → *Kural*) and connect thematically similar *Adhigarams* using embedding cosine similarity. Retrieval expands from high-scoring Kurals to their *Adhigaram*-cluster neighborhood, enforcing thematic consistency while keeping retrieval lightweight compared to LLM-driven GraphRAG-style construction.

d) *Evaluation of retrieval.*: We report Recall, Precision, MRR, and nDCG for each indexing strategy, holding prompts and models fixed across conditions.

IV. RESULTS AND ANALYSIS

We present results in three stages to directly address RQ1: (i) retrieval quality under different indexing strategies, (ii) end-to-end generation performance with RAG, and (iii) model behavior without retrieval.

A. Retrieval Performance Across Indexing Strategies

Table I reports retrieval metrics for the three indexing strategies, evaluated under identical prompts and candidate limits.

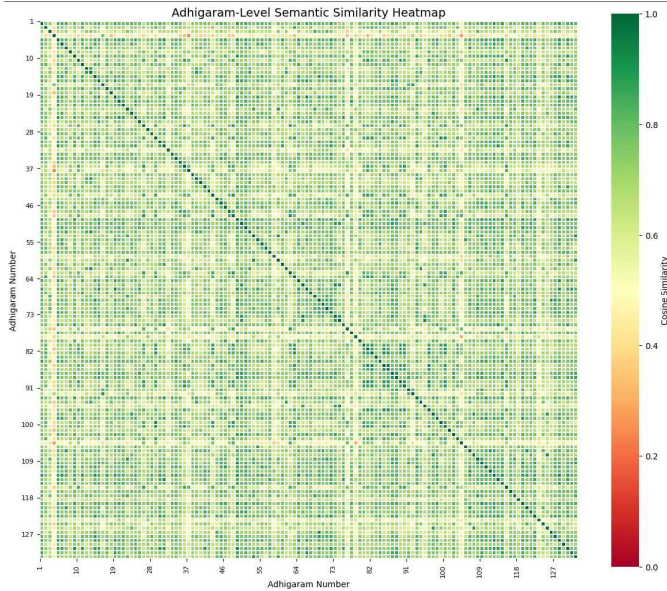


Fig. 3: Adhigaram-level semantic similarity heatmap.

TABLE I: Retrieval Performance Across Different Indexing Strategies

Indexing Strategy	Recall	Precision	MRR	nDCG
Kural-level indexing	0.009	0.005	0.026	0.256
Adhigaram-level indexing	0.408	0.245	0.626	0.705
Clustered Adhigaram-level indexing	0.999	0.777	0.796	0.986

Kural-level indexing performs poorly under strict verse-ID evaluation, despite using hybrid sparse+dense retrieval. This reflects the aphoristic and overlapping semantics of the corpus: multiple verses may plausibly address the same query, making exact verse recovery brittle. Moving to Adhigaram-level indexing substantially improves recall and ranking quality by shifting the retrieval unit from isolated couplets to thematically coherent chapters. The clustered Adhigaram strategy yields near-perfect recall and strong ranking metrics, demonstrating that enforcing hierarchical consistency during retrieval is critical for stable localization in culturally structured corpora.

B. End-to-End RAG Generation Performance

Table II reports end-to-end generation quality under RAG for large models and SLMs across indexing strategies.

Across all models, generation quality improves monotonically as retrieval structure becomes more coherent. Large models maintain relatively high semantic similarity even under weak retrieval, but still benefit from hierarchical indexing. In contrast, SLMs show sharp sensitivity to retrieval quality: under kural-level indexing, outputs are often generic or weakly grounded, whereas clustered Adhigaram-level retrieval yields substantial gains in both semantic metrics and retrieval-oriented measures. Notably, the fine-tuned 1B SLM approaches large-model performance under clustered retrieval, supporting the claim that explicit grounding can partially compensate for limited model capacity.

C. Generation Without Retrieval

To isolate the effect of parametric knowledge alone, Table III reports performance when retrieval is disabled.

Without retrieval, performance is dominated by model scale. Larger models achieve higher semantic similarity and moderate recall through memorized associations, while smaller models struggle to recover relevant verses or maintain consistent ranking. Fine-tuning improves stylistic consistency but does not substantially close the gap, confirming that retrieval—not adaptation—is the primary mechanism enabling faithful and attributable responses in this task.

D. Summary of Findings Relative to RQ1

Taken together, these results show that (i) retrieval structure is the dominant factor governing verse-level grounding, (ii) SLMs benefit disproportionately from well-organized evidence, and (iii) lightweight fine-tuning improves evidence usage but cannot substitute for retrieval. Under clustered hierarchical retrieval, SLMs can approach the performance of large models while operating under substantially lower computational constraints.

V. CONCLUSION

This paper evaluated whether small language models can provide faithful, attributable assistance over a culturally dense canonical text when explicit grounding is enforced. Using the Thirukkural as a testbed, we constructed a bilingual instruction corpus and a retrieval-augmented pipeline with bilingual reranking and hierarchical indexing aligned to the Paal–Iyal–Adhigaram structure. Experiments across base models, RAG-enabled models, and LoRA-adapted variants show a consistent pattern: retrieval organization is the dominant factor for both grounding and generation quality, and structured evidence can substantially narrow the gap between SLMs and much larger models.

Across indexing strategies, verse-level flat retrieval is unreliable under strict Kural-ID evaluation, while Adhigaram-level and clustered Adhigaram-level retrieval yields strong localization and more stable ranking. When paired with these retrieval settings, SLMs improve significantly in semantic metrics and citation-consistent explanations, demonstrating that faithful responses over small canonical corpora do not necessarily require large-scale parametric knowledge. Lightweight adaptation further improves response behavior, but it does not replace retrieval; it primarily improves evidence usage and stylistic consistency.

Overall, the results support the viability of deployable, frugal “cultural assistants” built from small models when the underlying canon is stable, bounded, and hierarchically structured. Future work should extend this framework to Tamil-first generation, richer commentary-aware evidence selection, and evaluation protocols that better capture valid multi-verse support while preserving strict attribution guarantees.

TABLE II: Performance of LLMs and SLMs under different indexing strategies in a RAG setup.

Model	Indexing Level	ROUGE-L	BERTScore	Precision	Recall	MRR
OpenAI/gpt-oss-20b	Kural	0.41	0.86	0.074	0.248	0.026
	Adhigaram	0.47	0.89	0.245	0.408	0.626
	Clustered	0.53	0.90	0.692	0.876	0.790
Gemini-2.5-flash	Kural	0.39	0.85	0.071	0.268	0.029
	Adhigaram	0.46	0.88	0.275	0.468	0.676
	Clustered	0.52	0.91	0.697	0.890	0.793
Perplexity-sonar	Kural	0.36	0.83	0.069	0.274	0.031
	Adhigaram	0.44	0.87	0.345	0.428	0.646
	Clustered	0.50	0.92	0.720	0.930	0.810
Gemma-3-270M	Kural	0.27	0.76	0.000	0.000	0.000
	Adhigaram	0.36	0.82	0.016	0.024	0.037
	Clustered	0.45	0.88	0.018	0.035	0.068
Gemma-3-1B	Kural	0.27	0.66	0.081	0.226	0.041
	Adhigaram	0.36	0.82	0.331	0.452	0.723
	Clustered	0.45	0.88	0.712	0.853	0.656
Gemma-3-270M (FT)	Kural	0.37	0.47	0.011	0.006	0.009
	Adhigaram	0.36	0.52	0.192	0.153	0.147
	Clustered	0.45	0.68	0.517	0.453	0.568
Gemma-3-1B (FT)	Kural	0.28	0.71	0.079	0.217	0.039
	Adhigaram	0.38	0.84	0.343	0.465	0.727
	Clustered	0.51	0.87	0.754	0.913	0.679

TABLE III: Evaluation Metrics for Different Models without RAG

Model	BERTScore	ROUGE-L	Recall	Precision	MRR
OpenAI/gpt-oss-20b	0.78	0.37	0.37	0.28	0.24
Gemini-2.5-flash	0.73	0.47	0.29	0.39	0.29
Perplexity-sonar	0.64	0.35	0.21	0.31	0.39
Gemma-3-270M	0.31	0.27	0.03	0.05	0.00
Gemma-3-1B	0.71	0.29	0.31	0.29	0.28
Gemma-3-270M (FT)	0.43	0.31	0.06	0.12	0.03
Gemma-3-1B (FT)	0.74	0.32	0.33	0.23	0.26

VI. LLM USAGE AND HUMAN CONTRIBUTIONS

We used large language models (LLMs) as *support tools* in limited roles that do not substitute the core scientific work. Specifically, LLMs were used for: (i) iterative refinement of paper phrasing and presentation (clarity edits, tightening paragraphs, and reorganizing sentences), and (ii) rapid prototyping (“vibe-coding”) of non-critical components such as plotting scripts, visualization templates, and minor statistics/formatting utilities.

The core research and development were performed by the authors. This includes: dataset design decisions (task definition, schema, split strategy, and holdout design), corpus curation and alignment, retrieval system design (indexing strategies, hybrid retrieval, reranking, and evidence formatting), implementation of grounding and attribution checks, training and LoRA configuration, experiment execution, and result interpretation. Human annotators performed semantic verification of question–Kural alignment, and author oversight was applied throughout model selection, evaluation design, and final reporting.

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